Agenda

- Earlier
  - Hypothesis testing
  - Parameters and Statistic
- Today
  - Parameter estimation
  - MLE

Alternative hypothesis	Evidence against H_0: $\theta > \theta_0$ provided by	P-value
H_1: $\theta > \theta_0$	too much bigger than θ_0 (i.e., $\hat{\Theta} - \theta_0$ too large)	$P = pr(T \geq t_0)$
H_1: $\theta < \theta_0$	too much smaller than θ_0 (i.e., $\hat{\Theta} - \theta_0$ too negative)	$P = pr(T \leq t_0)$
H_1: $\theta \neq \theta_0$	too far from θ_0 (i.e., $\lvert\hat{\Theta} - \theta_0\rvert$ too large)	$P = 2\ pr(T \geq \lvert t_0 \rvert)$
		where T ~ Student(df)

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Parameter and estimates

- A ***parameter*** is a numerical characteristic of a population or distribution. It often is used to "index" the distribution
- A **statistic** is a random variable and a function of the sample random variables $X_1,...,X_n$
- **Estimator**: A statistic $T(X_1,...,X_n)$ that is specifically being used as a guess for an unknown parameter θ
  - Example: The estimator $\bar{X}$
- After you actually observe, an ***estimate*** $\bar{x}$ is a realized quantity calculated from the data to estimate an unknown parameter
  - So we have done parameter estimation already! What's new?

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Parameters

- Bernoulli: $\theta = p$ (probability of success)
- Poisson: $\theta = \lambda$ (rate parameter)
- Uniform: θ is a vector of the upper and lower bounds
- Normal: θ is a vector of μ and σ^2
- More complex distributions such as the beta or the pareto distribution might well have more parameters
- Goal: How to find the parameters

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Context and workflow

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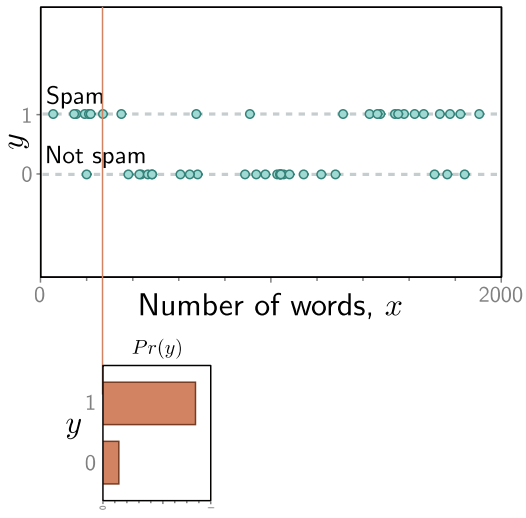
Context

- Step 1: “Real” business problems
  - Estimating the probability that at least two out of four servers will be alive next day
  - Is this email spam?
  - The probability that stock price will rise by 10% in the next week
  - The expected number of typos seen by a teacher in the next project report

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Context

- Step 2: Model the problem. Choose a functional form of the uncertainty.
  - Is this email spam?
  - Choose Bernoulli as the model, and find out “success” probability



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Context

- Step 1: Real business problems
  - Estimating the probability that at least two out of four servers will be alive next day
  - The probability that stock price will rise by 10% in the next week
  - The expected number of typos in project reports seen by a teacher in school
- Step 2: Model the problem. Choose a functional form of the uncertainty.
  - Bernoulli? Binomial? Gaussian? Poisson?

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Context

- Step 3: Collect training samples
  - Number of words in an email message
  - Sample server failure data observed over (say) n=3 days
  - Stock price change over the last 10 days
  - Number of typos in past reports
- Step 4: Estimate the unknown parameters using the training sample

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Review

- In the Cavendish experiment
  - What was the “real problem”
  - What was the model employed?
  - What kind of data collected?
  - How were the parameters reported? (Hint: use the word “interval” in your answer

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Fitting parameters to model

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Components of Solution

- Given: a density $f(x, \theta)$ with parameters θ
- Given: sample: $D = \{x_1, x_2, \ldots, x_I\}$
  - The i-th sample is a random variable X_i assumed to be iid as per $f(x, \theta)$
- Find θ
  - Since D is a finite sample, we cannot really know the actual θ. Best we can do is obtain an estimate of θ.
  - We will denote the estimate as $\hat{\theta}$

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Three methods

- Maximum likelihood
  - Sample D is all you have got
  - We get a *point* estimator $\hat{\Theta}$
- Maximum a posteriori and Bayesian approach
  - We also have some "prior" information

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Maximum Likelihood (MLE)

- As the name suggests: find the parameters under which the data $x_{1\ldots I}$ are most likely
- When viewed as a function of Θ $p(D|\Theta)$ is called the likelihood function and it is not necessarily a probability
- When viewed as a function of x_i, the conditional probability $p(D|\Theta)$ is a probability distribution that sums to 1

$$\hat{\boldsymbol{\theta}} = \arg\max_{\boldsymbol{\theta}} \Pr(\mathbf{x}_{1\ldots I}|\boldsymbol{\theta})$$
$$= \arg\max_{\boldsymbol{\theta}} \left[\prod_{i=1}^{I} \Pr(\mathbf{x}_i|\boldsymbol{\theta})\right]$$

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Bayes' Rule Terminology

Likelihood – propensity for observing a certain value of x given a certain value of y

Prior – what we know about y before seeing x

$$\Pr(y|x) = \frac{\Pr(x|y)\Pr(y)}{\int \Pr(x|y)\Pr(y)\,dy}$$

Posterior – what we know about y after seeing x

Evidence – a constant to ensure that the left-hand side is a valid distribution

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Maximum a posteriori (MAP) [1/2]

- As the name suggests we find the parameters which maximize the posterior probability $\Pr(\theta|x_{1\ldots I})$

$$\begin{aligned}\hat{\theta} &= \arg\max_{\theta} \Pr(\theta|\mathbf{x}_{1\ldots I}) \\ &= \arg\max_{\boldsymbol{\theta}} \left[\frac{\Pr(\mathbf{x}_{1\ldots I}|\boldsymbol{\theta})\Pr(\boldsymbol{\theta})}{\Pr(\mathbf{x}_{1\ldots I})}\right] \\ &= \arg\max_{\boldsymbol{\theta}} \left[\frac{\prod_{i=1}^{I}\Pr(\mathbf{x}_i|\boldsymbol{\theta})\Pr(\boldsymbol{\theta})}{\Pr(\mathbf{x}_{1\ldots I})}\right]\end{aligned}$$

Again we have assumed that data was independent

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Maximum a posteriori (MAP) [2/2]

Maximize the posterior probability $\Pr(\theta|x_{1\ldots I})$

$$\hat{\theta} = \arg\max_{\boldsymbol{\theta}} \left[\frac{\prod_{i=1}^{I}\Pr(\mathbf{x}_i|\boldsymbol{\theta})\Pr(\boldsymbol{\theta})}{\Pr(\mathbf{x}_{1\ldots I})}\right]$$

Point estimate

Since the denominator doesn't depend on the parameters we can instead maximize

$$\hat{\theta} = \arg\max_{\theta} \left[\prod_{i=1}^{I}\Pr(\mathbf{x}_i \mid \theta)\Pr(\theta)\right]$$

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Bayesian Approach [1/2]

Compute the posterior *distribution* over possible parameter values using Bayes' rule:

$$\Pr(\boldsymbol{\theta} \mid \mathbf{x}_{1\ldots I}) = \frac{\prod_{i=1}^{I} \Pr(\mathbf{x}_i \mid \boldsymbol{\theta}) \Pr(\boldsymbol{\theta})}{\Pr(\mathbf{x}_{1\ldots I})}$$

Not a point estimate

Principle: why pick one set of parameters? There are many values that could have explained the data. Try to capture all possibilities

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Predictive densities for 3 methods

Maximum likelihood:
Evaluate new data point $\boldsymbol{x}^*$ under probability distribution with ML parameters $\Pr(\mathrm{x}^* | \widehat{\boldsymbol{\theta}})$

Maximum a posteriori:
Evaluate new data point $\boldsymbol{x}^*$ under probability distribution with MAP parameters $\Pr(\mathrm{x}^* | \widehat{\boldsymbol{\theta}})$

Bayesian:
Calculate weighted sum of predictions from all possible values of parameters

$$\Pr(\mathbf{x}^* \mid \mathbf{x}_{1\ldots I}) = \int \Pr(\mathbf{x}^* \mid \boldsymbol{\theta}) \Pr(\boldsymbol{\theta} \mid \mathbf{x}_{1\ldots I}) \, d\boldsymbol{\theta}$$

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